# Chapter 6. Sequences and Series of Functions

Uniform convergence, differentiation and integration of limits, double series, power series, trigonometric series

## Contents


**6.1 Uniform convergence** **2**

**6.2 Differentiation and integration of limits** **2**

**6.3 Double sequences and double series** **3**

**6.4 Power series** **3**

**6.5 Trigonometric series** **4**

When can limits of functions be exchanged with continuity, derivatives, integrals and other limits? The answer, in most cases, is uniform convergence.

## 6.1 Uniform convergence

Let $f_n, f : X \to \mathbb{R}^m$ ($X \subset \mathbb{R}^k$). $f_n \to f$ **pointwise** if $f_n(x) \to f(x)$ for each $x$, and **uniformly** if $\|f_n - f\|_\infty = \sup_{x\in X}\|f_n(x) - f(x)\| \to 0$, i.e. $N$ can be chosen independently of $x$.

**Example 6.1**

(a) $x^n$ on $[0,1]$ converges pointwise to the function equal to 0 on $[0,1)$ and 1 at 1: continuous functions with a discontinuous limit, so the convergence is not uniform. On $[0,r]$ with $r<1$ it is uniform.
(b) $f_n(x) = nx(1-x^2)^n$ on $[0,1]$ tends to 0 pointwise, but $\int_0^1 f_n = \frac{n}{2n+2} \to \frac{1}{2}$.

**Theorem 6.1**

A uniform limit of continuous functions is continuous.

**Proof.** $\|f(x) - f(x_0)\| \le \|f(x) - f_n(x)\| + \|f_n(x) - f_n(x_0)\| + \|f_n(x_0) - f(x_0)\|$: choose $n$ with $\|f_n - f\|_\infty < \epsilon$, then $\delta$ from the continuity of $f_n$ at $x_0$ (the "$\epsilon/3$ argument"). ■

**Theorem 6.2 (Uniform Cauchy criterion)**

$(f_n)$ converges uniformly on $X$ iff for every $\epsilon > 0$ there is $N$ with $\|f_n - f_m\|_\infty < \epsilon$ for $m, n \ge N$.

**Proof.** Sufficiency: for each $x$ the sequence is Cauchy, so it converges to some $f(x)$; letting $m \to \infty$ in $\|f_n(x) - f_m(x)\| < \epsilon$ gives $\|f_n - f\|_\infty \le \epsilon$ for $n \ge N$. ■

**Corollary 6.3 (Weierstrass M-test)**

If $|f_n(x)| \le M_n$ on $X$ and $\sum M_n < \infty$, then $\sum f_n$ converges absolutely and uniformly on $X$.

**Example 6.2**

(a) $\sum \frac{\sin nx}{n^2}$ converges uniformly on $\mathbb{R}$ to a continuous function.
(b) (A continuous nowhere differentiable function.) Let $\phi(x)$ be the distance from $x$ to the nearest integer and $f(x) = \sum_{k\ge 0}(3/4)^k \phi(4^k x)$. The M-test gives continuity. At any $x$, choosing $h_m = \pm\frac{1}{2}4^{-m}$ so that no integer lies between $4^m x$ and $4^m(x+h_m)$ makes $|f(x+h_m) - f(x)|/|h_m| \ge 3^m - \sum_{k<m} 3^k = \frac{3^m+1}{2} \to \infty$, so $f$ is nowhere differentiable.

## 6.2 Differentiation and integration of limits

**Theorem 6.4**

Let $f_n$ be differentiable on $[a,b]$, $(f_n(x_0))$ converge for some $x_0$, and $(f_n')$ converge uniformly. Then $(f_n)$ converges uniformly to a differentiable $f$ with $f' = \lim f_n'$.

**Proof.** The mean value theorem applied to $f_n - f_m$ gives $|(f_n - f_m)(x) - (f_n - f_m)(t)| \le \|f_n' - f_m'\|_\infty |x - t|$. With $t = x_0$ this makes $(f_n)$ uniformly Cauchy. Fix $x$ and put $\phi_n(t) = \frac{f_n(t) - f_n(x)}{t - x}$,

$\phi(t) = \frac{f(t)-f(x)}{t-x}$; the same estimate shows $\phi_n \to \phi$ uniformly on $t \neq x$, and $\lim_{t\to x} \phi_n(t) = f_n'(x)$. Exchanging the two limits (allowed by uniformity, as in Theorem 6.1) gives $f'(x) = \lim f_n'(x)$. ■

Uniform convergence of $f_n$ alone is not enough: $\frac{\sin nx}{\sqrt{n}} \to 0$ uniformly but $\sqrt{n}\cos nx$ diverges.

**Proposition 6.5**

If $f_n \in \mathcal{R}[a,b]$ and $f_n \to f$ uniformly, then $f \in \mathcal{R}[a,b]$ and $\int_a^b f_n \to \int_a^b f$.

**Proof.** With $\epsilon_n = \|f_n - f\|_\infty$, $f_n - \epsilon_n \le f \le f_n + \epsilon_n$, so $\overline{\int} f - \underline{\int} f \le 2\epsilon_n(b-a) \to 0$ and $|\int f - \int f_n| \le \epsilon_n(b-a)$. ■

Hence $\sum f_n$ may be integrated term by term when it converges uniformly. Example 6.1(b) shows what goes wrong without uniformity; Chapter 10 gives far more flexible conditions (dominated and monotone convergence).

## 6.3 Double sequences and double series

**Proposition 6.6 (Exchanging limits)**

Let $(a_{mn})$ be a double sequence such that $a_{mn} \to b_m$ as $n \to \infty$ uniformly in $m$, and $a_{mn} \to c_n$ as $m \to \infty$ for each $n$. Then $\lim_m b_m$ and $\lim_n c_n$ exist and are equal.

**Proof.** Uniformity in $m$ makes $(c_n)$ Cauchy: $|a_{mn} - a_{mn'}| < \epsilon$ for $n, n' \ge N$ and all $m$; let $m \to \infty$. With $c = \lim c_n$, $|b_m - c| \le |b_m - a_{mn}| + |a_{mn} - c_n| + |c_n - c|$; fix $n$ large to control the first and third terms, then let $m \to \infty$. ■

This is the abstract form of Theorem 6.1 (take $a_{mn} = f_n(x_m)$). For double series the cleanest result concerns nonnegative terms.

**Proposition 6.7 (Double series)**

If $a_{mn} \ge 0$, then $\sum_m \sum_n a_{mn} = \sum_n \sum_m a_{mn}$ (possibly $+\infty$), and both equal the sum over $\mathbb{N}^2$ in any order. If $\sum_m \sum_n |a_{mn}| < \infty$, the same holds for real or complex $a_{mn}$.

**Proof.** For finite $M, N$, $\sum_{m\le M}\sum_{n\le N} a_{mn} \le \sum_n \sum_m a_{mn}$; let $N \to \infty$, then $M \to \infty$; and symmetrically. Every finite partial sum over $\mathbb{N}^2$ is bounded by either iterated sum, and conversely. For absolutely summable families, split into positive and negative (real and imaginary) parts. ■

Without absolute convergence the iterated sums can differ: with $a_{mm} = 1$, $a_{m,m+1} = -1$ and $0$ otherwise, $\sum_m \sum_n = 0$ but $\sum_n \sum_m = 1$.

## 6.4 Power series

**Theorem 6.8 (Cauchy–Hadamard)**

Let $A = \limsup |a_n|^{1/n}$ and $R = 1/A$ ($R = \infty$ if $A = 0$, $R = 0$ if $A = \infty$). Then $\sum a_n x^n$ converges absolutely for $|x| < R$, diverges for $|x| > R$, and converges uniformly on $[-r, r]$ for every $r < R$.

**Proof.** Apply the root test: $\limsup |a_n x^n|^{1/n} = A|x|$. For $|x| \le r < R$, the M-test with $M_n = |a_n| r_1^n$, $r < r_1 < R$, gives uniformity. ■

So $f(x)=\sum a_n x^n$ is continuous on $(-R,R)$. Since $\limsup(n|a_n|)^{1/n}=A$, the differentiated series has the same radius, and Theorem 6.4 gives:

**Theorem 6.9**

$f(x)=\sum a_n x^n$ is infinitely differentiable on $(-R,R)$ with $f'(x)=\sum n a_n x^{n-1}$; in particular $a_n=f^{(n)}(0)/n!$. Moreover, for $|c|<R$, $f(x)=\sum_k \frac{f^{(k)}(c)}{k!}(x-c)^k$ for $|x-c|<R-|c|$.

**Proof.** The re-expansion: $\sum_n a_n(c+(x-c))^n=\sum_n\sum_k \binom{n}{k}a_n c^{n-k}(x-c)^k$, and the double series converges absolutely when $|c|+|x-c|<R$, so Proposition 6.7 allows the order to be exchanged. ■

**Proposition 6.10 (Cauchy product)**

If $\sum a_n=A$ converges absolutely and $\sum b_n=B$ converges, then $\sum c_n$, $c_n=\sum_{k=0}^n a_k b_{n-k}$, converges to $AB$ (Mertens).

**Proof.** With $B_n=\sum_{k\le n} b_k$ and $\beta_n=B_n-B\to 0$, $\sum_{k\le n} c_k=\sum_{k\le n} a_k B_{n-k}=A_nB+\sum_{k\le n} a_k\beta_{n-k}$, and the last sum tends to 0 by splitting at a large index and using $\sum|a_k|<\infty$. ■

**Theorem 6.11 (Abel)**

If $\sum a_n$ converges, then $\sum a_n x^n$ converges uniformly on $[0,1]$; hence $\lim_{x\to 1-}\sum a_n x^n=\sum a_n$.

**Proof.** Let $r_n=\sum_{k>n} a_k\to 0$. Summation by parts gives $\sum_{k=m}^n a_k x^k=r_{m-1}x^m-r_n x^n-\sum_{k=m}^{n-1} r_k(x^k-x^{k+1})$, which is at most $3\sup_{k\ge m-1}|r_k|$ for $0\le x\le 1$, because $\sum(x^k-x^{k+1})\le 1$. ■

**Example 6.3**

(a) $\log(1+x)=\sum_{n\ge 1}(-1)^{n+1}x^n/n$ on $(-1,1)$ (integrate $1/(1+x)$ term by term); Abel's theorem gives $\log 2=1-\frac12+\frac13-\cdots$. Similarly $\frac{\pi}{4}=1-\frac13+\frac15-\cdots$ from arctan.
(b) As a consequence of Abel's theorem, if $\sum a_n$, $\sum b_n$ and their Cauchy product $\sum c_n$ all converge, then $\sum c_n=(\sum a_n)(\sum b_n)$: apply Abel's theorem to $\sum c_n x^n=(\sum a_n x^n)(\sum b_n x^n)$, valid for $|x|<1$.
(c) The converse of Abel's theorem fails: $\sum(-1)^n x^n\to\frac12$ as $x\to 1-$ but $\sum(-1)^n$ diverges.

## 6.5 Trigonometric series

For complex vectors we use $\langle z,w\rangle=\sum z_i\overline{w_i}$; the Cauchy–Schwarz inequality $|\langle z,w\rangle|\le\|z\|\|w\|$ holds in any complex inner product space, with the same proof (minimize $\|z-tw\|^2$ over complex $t$).

**Theorem 6.12 (Dirichlet's test)**

Let $f_n$ be complex functions on $X$ whose partial sums $F_n=\sum_{k\le n} f_k$ are uniformly bounded, and let $g_n$ be real with $g_1\ge g_2\ge\cdots\ge 0$ and $g_n\to 0$ uniformly. Then $\sum f_n g_n$ converges uniformly on $X$.

**Proof.** Summation by parts: $\sum_{k=m}^n f_k g_k=F_n g_n-F_{m-1}g_m+\sum_{k=m}^{n-1}F_k(g_k-g_{k+1})$. If $|F_k|\le M$, the absolute value is at most $M\{g_n+g_m+(g_m-g_n)\}=2Mg_m\to 0$ uniformly; apply the uniform Cauchy criterion. ■

In particular, if $\sum a_n$ has bounded partial sums and $b_n\downarrow 0$, then $\sum a_n b_n$ converges (this contains the alternating series test).

**Theorem 6.13**

If $b_n \downarrow 0$, then $\sum b_n e^{inx}$, $\sum b_n \cos nx$ and $\sum b_n \sin nx$ converge uniformly on $\{\delta \le |x| \le \pi\}$ for every $\delta \in (0, \pi)$. In particular $\sum b_n \sin nx$ converges for every $x$.

**Proof.** $\left|\sum_{k \le n} e^{ikx}\right| = \left|\frac{e^{i(n+1)x} - e^{ix}}{e^{ix} - 1}\right| \le \frac{1}{|\sin(x/2)|} \le \frac{1}{\sin(\delta/2)}$; apply Dirichlet's test. At $x \in \pi\mathbb{Z}$ the sine series is 0. ∎

More generally the conclusion holds if $b_n \to 0$ and $\sum |b_n - b_{n-1}| < \infty$ (the same proof). For instance $\sum \frac{\sin nx}{n}$ converges everywhere (to $\frac{\pi - x}{2}$ on $(0, 2\pi)$, Chapter 9), but not uniformly near 0.

**Theorem 6.14 (Abel's test)**

If $\sum f_n$ converges uniformly on $X$ and $(g_n)$ are real functions, uniformly bounded and monotone in $n$ for each $x$, then $\sum f_n g_n$ converges uniformly.

**Proof.** Apply summation by parts with the tails $r_n = \sum_{k > n} f_k$, which are uniformly small, as in the proof of Theorem 6.11. ∎